MILITARY SPECIFICATION

OXYGEN SYSTEM, SURVIVAL CONTAINER-OXYGEN KIT, GENERAL SPECIFICATION FOR

This specification is approved fo use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for an oxygen system to be installed in survival containers of aircraft requiring the use of pressure coveralls and helmets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-T-60	Tape, Packaging, Waterproof
BB-N-411	Nitrogen, Technical
PPP-B-636	Box, Shipping, Fiberboard
BB-A-1034	Air, Compressed, For Breathing Purposes

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-W-5088	Wiring, Aerospace Vehicle
MIL-T-6117	Terminal, Cable Assemblies, Swaged Type
MIL-W-6858	Welding, Resistance, Spot And Seam
MIL-G-7601	Gage, Emergency Oxygen, Type L-2
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics/MMEDO, Tinker AFB OK 73145-5990, by using the self-addressed Standardizatin Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-8516	Sealing Compound, Polysulfide Rubber, Electric Connectors And Electric Systems, Chemically Cured
MIL-W-8611	Welding, Metal Arc And Gas, Steels, And Corrosion And Heat Resisting Alloys, Process For
MIL-R-25572	Regulator, Oxygen, Barometric Demand, General Specification For
MIL-R-25575	Valve, Regulations, Fluid Pressure, Aircraft Oxygen System, General Specifications For
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-A-27471	Adapter, Pressure-reducer, In Line CRU-43A/A
MIL-S-83427	Survival Kit Containers, CNU-129 (V) P and CNU-130/P

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-781	Reliability Tests Exponetial Distribution
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-831	Test Reports Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-2175	Casting, Classification And Inspection Of
MS20995	Wire, Safety Or Lock
MS22066	Valve, Check, Oxygen Charging
MS26545	Cylinder Compressed Gas, Nonshatterable
MS33540	Safety Wiring And Cotter Pinning, General Pratices For
MS33656	Fitting End, Standard Dimensions For Flared Tube Connection And Gasket Seal
MS33658	Fitting End, Hose Connection, Standard Dimension For

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Air Force

67D25374	Personal Leads Assembly
67E25232	Oxygen System, Installation of

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article inspection (see 4.4 and 6.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143, except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS and AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. Any material that deteriorates easily or is otherwise easily affected adversely by continued service use with oxygen shall not be used. The materials shall be entirely suitable for the purpose intended. Materials that will come in contact with oxygen shall be entirely free from oil, grease, or any other combustible material.

3.3.1 Nonferrous materials. Nonferrous materials shall be used for all parts of the oxygen system except where ferrous materials are essential.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion caused by fuels, salt spray, or any other atmospheric condition likely to be met in storage or normal service.

3.3.2.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless they are suitably protected against electrolytic corrosion. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Nonmetallic materials. Nometallic materials used shall be uniform in structure, free from flaws, durable, adequate in strength, nonbrittle, tear resistant and nontacky. Nonmetallic materials used in the oxygen unit shall be nontoxic and shall not contain objectionable odors. Natural rubber materials shall not be used. Elastomers used in the oxygen unit shall be of silicone rubber.

3.3.4 Protective treatment. The materials used in the construction of the oxygen system that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.4.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

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3.4 Design and construction. The survival container oxygen system shall consist basically of a regulator, reducers, manifold, kit-to-man disconnect, personal leads, aircraft to oxygen system disconnect, pressure gage, recharging valve, manual and automatic activation mechanism, oxygen cylinders, electrical connectors, hoses, tubing, fittings, etc. necessary to regulate the 70 pounds per square inch (psi) aircraft oxygen supply.

3.4.1 Oxygen regulator and press-to-test mechanism. The system shall incorporate an oxygen regulator and press-to-test mechanism conforming to MIL-R-25572. The press-to-test mechanism shall be located as shown on AF drawing 67E25232.

3.4.2 Reducer. The system shall incorporate a reducer conforming to MIL-R-25575 for reducing the pressure of the emergency oxygen supply.

3.4.3 Manifold. The manifold shall contain the connection for the press-to-test line and shall integrate with the mounting plate to form a locking device for the oxygen system in accordance with AF drawing 67E25232.

3.4.4 Kit-to-man disconnect. The kit-to-man disconnect shall consist of a lower disconnect conforming to AF drawing 67E25232 and shall mate with an upper disconnect per AF drawing 67D25374 for use with the intended full pressure or partial pressure suit. The kit-to-man disconnect shall convey the following services from the oxygen package to the personal leads: Capstan pressure, breathing and chest bladder oxygen, communications, visor heat, a single 70 psi (nominal) oxygen supply, and an emergency oxygen manual release mechanism for the applicable full or partial pressure suit. The 70-psi oxygen supply line shall contain a check valve which shall open only when the applicable pressure suit upper disconnect is inserted and shall, at all other times, stay close and leak tight.

3.4.5 Personal leads. The personal leads assmebly shall conform to AF drawing 67D25374 for use with the applicable pressure suit and helmet.

3.4.5.1 Pressure reducer. A pressure reducer adapter conforming to MIL-A-27471 shall be incorporated within the system when specified in the contract. The pressure reducer shall be required in aircraft utilizing personal leads conforming to AF drawing 67D25374 for adapting the system for use with low altitude oxygen masks.

3.4.6 Aircraft-to-oxygen system disconnect. The aircraft-to-oxygen system disconnect shall separate and actuate the oxygen supply when a pull is applied to the emergency oxygen activation cable in any direction within 10 degrees of the line of travel of the parting disconnect. The disconnect shall incorporate a standard male, 5/16 inch flare fitting end, MS33656, for connecting the ship's oxygen supply; a mechanical mechanism operated only by the ship's cable shall prevent accidental actuation. The locking mechanism shall have an indicator to show the full locked position. Provision shall be made for manual separation for maintenance purposes. The actuation apparatus shall be locked in the actuated condition until the disconnect is re-engaged. Both halves of the aircraft-to-oxygen system disconnect shall contain check valves in the 70 psi (nominal) oxygen supply line which shall open only when the aircraft portion of the disconnect is inserted and shall, at all other times, stay closed and leak tight.

3.4.7 Pressure gage. The system shall incorporated a readily visible (pre-flight only) pressure gage conforming to MIL-G-7601 for the emergency oxygen supply.

3.4.8 Recharging mechanism. The system shall incorporate a fitting conforming to MS22066 for readily recharging the emergency oxygen supply.

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3.4.9 Manual activation mechanism "green apple". The system shall incorporate a manual activation mechanism "green apple" for manual activation of the emergency oxygen supply without ejection. This mechanism shall be in accordance with the applicable oxygen system drawing. The emergency oxygen, after going through the reducer, shall tee into the regulator by-pass line to adequately service the full pressure suit as well as the partial pressure suit.

3.4.10 Oxygen cylinders. The oxygen system requires two nonshatterable oxygen cylinders suitable for installation in the aft compartment of the survival container. The cylinders shall be Type A, Class 2, with a 25 cubic inch minimum capacity each in accordance with MS26545.

3.4.11 Electrical connectors. The electrical connectors shall be potted with sealing compound meeting the requirements of MIL-S-8516 or equivalent. Adequate strain relief shall be provided for wires where entering the connectors.

3.4.12 Miscellaneous components. The system shall incorporate necessary tubing, hoses, fittings, wires, cables, etc., for use with the applicable partial and full pressure suits, helmets, and communication systems. The system shall also incorporate a regulator-by-pass tubing and, when specified in the contract or order, the necessary personal leads for use with the intended pressure flying suits. The system shall incorporate necessary disconnects, check valves, and other necessary components to permit normal, emergency, or ejection operation with the applicable suits and helmets.

3.4.13 Loss of emergency oxygen. The system shall incorporate a suitable mechanism to prevent loss of emergency oxygen due to bailout, depletion of the aircraft's oxygen supply, or rupture of the aircraft's oxygen supply system.

3.4.14 Plugs. All inlets and outlets of the oxygen system shall be closed with caps or plugs to prevent dust and foreign material from entering during shipment or storage.

3.4.15 End fitting hose connectors. Unless otherwise specified in applicable Air Force drawings, all hose connection end fittings for low pressure oxygen personal leads assemblies shall be designed in accordance with MS33658 with the following exceptions:

a. Fitting length shall be 1.140 inch $\pm$ 0.047.

b. Inner diameter of the 3/8 inch fitting end shall be 0.302 inch minimum.

c. Change the .093 dimension to 0.140 inch, plus or minus 0.050.

d. Change the .250 dimension to 0.343 maximum.

3.5 Design objectives. The system, when properly integrated into a survival container per MIL-S-83427 shall adequately furnish oxygen and communications to the intended pressure suits, helmets and communication systems, under the conditions of pressure, temperature, and use which will be experienced in the applicable weapon systems.

3.5.1 Integration with survival container. The oxygen system shall be designed to integrate with the applicable survival container in accordance with MIL-S-83427 to allow the following operations:

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a. Ejection with automatic activation of emergency oxygen supply and subsequent manual release of the kit from the man. This operation shall be accomplished in the following steps:

(1) Firing seat shall automatically activate oxygen.

(2) Manual actuation by pulling the kit handle or automatic actuation of the kit hardware shall simultaneously disconnect personal leads at the kit and separate the kit from the man. Separation of the personal leads will be accomplished by causing the actuator lever of AF drawing 67E25232 to move through a distance of 3/8 inch, minimum, at the actuator cable. Sufficient over-travel shall be provided to prevent jamming.

b. Emergency abandonment of aircraft on the ground or in the water (same as a.(2)).

c. Normal egress from the aircraft. The helmet hose and wiring shall be disconnected at the personal disconnects and the suit hose shall be disconnected at the suit connector.

d. Removal of the kit from the aircraft for servicing, packing, etc. Provisions shall be made for disconnection of all attachments (emergency oxygen cable, etc.) to the seat and to the aircraft without the use of special tools and without repacking the container.

e. Removal of the oxygen system from the container for testing, replacemnt, etc. A suitable lock shall be provided to hold the oxygen system so that it may be installed and removed without tools of any kind.

3.5.2 Reliability. The oxygen system shall have a mean-cycle-between-failure (MCBF) of 100 cycles at a 90 percent confidence level. A cycle shall consist of one operation as specified in 3.5.1 subparagraph a.

3.5.3 Longevity. The oxygen system shall have an operating life span (equipment longevity is defined in MIL-STD-781) of not less than 300 cycles before wearout failures occur or the equipment consistently fails to meet the specified MCBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the contracting activity.

3.6 Performance. The oxygen system shall perform satisfactorily when subjected to the conditions specified in test methods. 4.6.

3.7 Weight. Components shall be made of the lightest possible materials consistent with configuration, strength, and other requirements specified herein.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Workmanship. The oxygen system shall be manufactured in a thoroughly workmanlike manner. Particular attention shall be paid to accuracy of dimensions, radii of fillets, marking of parts and assemblies, thoroughness of welding, brazing, painting, riveting, swaging, soldering, potting, alignment of parts, tightness of screws and bolts, etc. The oxygen system shall be built to withstand the strains,

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jars, and other conditions incidental to shipping, storage, installation and service without failure. Welding, riveting, or other permanent methods of assembly may be used provided that disassembly or reassembly of the component for inspection or overhaul purposes may be accomplished rapidly, safely, and without damage to the assembly. Operation after assembly shall not be dependent upon selective fit of any or all component parts.

3.9.1 Riveting. The riveting operation shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.9.2 Welding. Welding shall be in accordance with MIL-W-8611 or MIL-W-6858 as applicable and shall be carefully performed to insure that all welded joints shall have proper penetration, proper bead, and proper strength.

3.9.3 Wiring. Wiring shall be accomplished in accordance with MIL-W-5088. All electrical wiring shall be installed with liberal bends to prevent undue strain on insulation and wires. Wires shall be protected from all moving parts.

3.9.4 Castings. Castings shall be of high quality, clean, sound, free of holes, porosity, cracks, and any other defects. Castings shall be designed and inspected in accordance with MIL-STD-2175. Castings shall not be used in applications where failure could result in loss or failure of oxygen system during ejection and descent.

3.9.5 Soldering and brazing. Soldering and brazing shall be done in accordance with current applicable specifications and shall be carefully performed to insure that all joints have proper penetration, proper bead, and proper strength.

3.9.6 Swaging. Swaging of cable fittings shall be done in accordance with MIL-T-6117 and shall be carefully performed. Each swaging operation shall be proof tested to 100 pounds or three times the maximum service load, whichever is greater.

3.9.7 Alignment. The alignment of fitting and mating surfaces shall be accurate to a degree that will permit the proper functioning of the unit within the environmental conditions expected during the service life of the unit.

3.9.8 Lubrication. All parts of the oxygen system shall be entirely free from oil, grease, or other combustible materials. Lubricants used in the oxygen system which may be subject to exposure to oxygen shall be of the type approved for this purpose by the contracting activity.

3.9.9 Threads. Except for permanently installed pipe threaded plugs, or as necessitated by functional purposes and unless otherwise specified in the applicable documents, only straight threads conforming to MIL-S-7742, National Fine Thread Series, Class 3 (NF-3) or Unified Thread Series, Class 3A or 3B shall be used.

3.9.10 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or distortion to the screw, bolt, nut, or components being secured by the screw or bolt.

3.9.11 Safetying. All threaded parts shall be positively locked or safetied by safety wire, self locking nuts, or any other approved methods. Safety wire shall be applied in accordance with the practice outlined in MS33540, and shall conform to MS20995.

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3.9.12 Cleaning. The oxygen system shall be thoroughly cleaned. All oils, grease, and combustible or foreign material shall be removed during and after final assembly.

3.9.13 Protection. The oxygen systems shall be protected against the entry of all foreign materials.

3.10 Identification of product. All assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Name plate. A nameplate containing the following information shall be securely attached to the oxygen system in a prominently legible position.

OXYGEN SYSTEM, SURVIVAL CONTAINER-OXYGEN KIT
Specification MIL-O-27335A
Manufacturer's Part No.
Manufacturer's Serial No.
Federal Stock No.
Contract No.
Manufacturer's Name or Trade Mark (omit address)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection and testing of the oxygen system shall be classified as follows:

a. Quality, conformance inspection (See 4.3)

b. First article inspection (See 4.4)

4.3 Quality conformance inspection.

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4.3.1 Inspection of products for delivery. Inspection of products ready for delivery shall consist of group A and group B inspections; reference tables I and II.

4.3.1.1 Inspection lot. A lot shall consist of oxygen systems manufactured under essentially the same conditions and at essentially the same time and shall not exceed 100.

4.3.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table I and shall be made on the same set of sample units.

4.3.1.2.1 Sampling plan. Group A tests shall be performed on 100 percent of the products supplied under this specification. Individual components which fail the required tests shall be removed from the lot. Lots having more than 5% rejects shall not be furnished on contracts. Evidence of noncompliance shall be cause for rejection.

4.3.1.2.1.2 Rejected logs (Group A). Rejected inspection lots may be resubmitted for Government acceptance only if the manufacturer performs 100 percent inspection on components of the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units and resubmits the lot for quality conformance inspection. Resubmitted lots shall be kept separate from new lots, and shall be clearly identified as resubmitted lots.

4.3.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table III in the order shown. They shall be performed on sample units that have been subjected to and have passed the group A inspection.

4.3.1.3.1 Sampling plan. Two oxygen systems shall be selected at random from each lot or fraction thereof which passed the inspections of table I (Group A) shall then be subjected to the tests listed in table II (Group B).

4.3.1.3.2 Rejected lots (Group B). If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.3.1.3.3 Group A tests may continue. For production reasons, Group A tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.1.3.4 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, provided they are within resistance and capacitance tolerance and meet requirements for visual and mechanical inspection.

4.4 First article.

4.4.1 First article inspection. First article inspection shall apply to samples and shall include all inspections listed in tables I, II & III. When specified in the contract or order, a test report shall be prepared (see 6.2.2).

4.4.2 Test samples. The following shall be furnished to the contracting activity:

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a. Two samples which were tested.

4.4.2.2 Disposition of samples. Sample units which have been subjected to group inspection shall not be delivered on the contract or purchase order.

4.4.2.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.4.3 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing, and marking or shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

4.5 Inspection conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees, plus or minus 18 degrees Fahrenheit, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.5.2 Gas. The high-pressure gas used in testing the oxygen system components shall be oxygen conforming to Type I of MIL-O-27210; water-pumped nitrogen conforming to Type 1, Grade B, Class 1 of BB-N-411; or compressed air conforming to Source 1, Grade 1 or Source II, Grade C of BB-A-1034. If nitrogen or air is used, appropriate density correction factors shall be applied to the flowmeter used, not only to correct the effect on the meter, but also on the performance of the valve with lower density gas. Only oxygen shall be used to charge the complete system.

4.6 Test methods.

4.6.1 Examination of product. Each oxygen system shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, design objectives and identification of product.

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4.6.2 Kit-to-man disconnect. The partial pressure suit and helmet oxygen connectors shall be plugged and the oxygen system shall be supplied with 100 psi oxygen through the aircraft supply line in a simulated installation. The cylinder shall be charged with oxygen to 1800 psi. The maximum force required by the container release handle to actuate the disconnect shall not exceed 5 pounds. This test shall be repeated using the full pressure suit lead.

4.6.3 Leakage, kit-to-man disconnect. The suit and helmet oxygen connectors shall be plugged and oxygen shall be admitted to the disconnect outlets from the regulator at a pressure of 15 psi for the suit port and 3 psi for the helmet port. The regulator-by-pass line shall be charged to 120 psi. Leakage of the kit-to-man disconnect shall not exceed 0.05 liter per minute. The test shall be repeated using any other type pressure suit leads specified in the contract or order.

4.6.4 Operation "green apple". With ports blocked off, the "green apple" pull cable shall be operated after resetting the pressure reducer and recharging the oxygen system. The pull cable shall actuate the emergency supply and shall not jam. Leakage through the aircraft to oxygen system disconnect of more than 0.1 liters per minute or jamming of the pull cable shall be cause for rejection. The pull required on the "green apple" pull cable to actuate the emergency supply shall be 20 pounds, plus or minus 5 pounds.

4.6.5 Aircraft-to-oxygen system disconnect. The complete oxygen system shall be installed as in a container installation (see AF Drawing 67E25232) and shall be subjected to five (5) successive simulated ejections with automatic separation of the aircraft-to-oxygen system disconnect. This test shall be conducted with the emergency oxygen system charged to 1800 psi, the aircraft supply at 100 psi, the pressure reducer set and the suit and helmet connectors blocked off. The force required to separate the aircraft-to-oxygen system disconnect shall be 40 pounds, plus or minus 20 pounds, when the cable is pulled at any angle within 10 degrees of line of travel of the parting disconnect.

4.6.6 Pressure drop-suit operation. The complete oxygen system with full pressure suit personal leads installed shall be tested for flow resistance through the full pressure suit section. The static pressure drop through the oxygen system shall not exceed 10 pounds per square inch with 200 liters per minute flow at 70 pounds per square inch gage inlet pressure. Leakage through either portion of the disconnects shall not exceed 0.1 liters per minute.

4.6.7 Leakage - oxygen systems. Each oxygen system shall be leak tight as determined by the following test. The cylinders shall be charged with oxygen to 2100 psi and the position of the gage indicator marked after the temperature of the assembly is stabilized. After 24 hours of storage, any change in the position of the gage indicator shall be cause of rejection.

4.6.8 Electrical continuity. Electrical circuits shall be checked for continuity and correctness of wiring by a continuity tester.

4.6.9 Dielectric strength. The electrical circuit shall be tested by applying an alternating current voltage of 500 volts (rms) for a period of not less than one (1) minute between each terminal and all other terminals. There shall be no arc-over or breakdown of the insulation material.

4.6.10 Leakage, oxygen manifold. The press-to-test port of the oxygen manifold shall be pressurized to 3 pounds per square inch. Leakage shall not exceed 0.05 liters per minute.

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4.6.11 Proof load, disconnect cable. The cable assembly which is a component part of the oxygen system-to-man disconnect shall be proof loaded to 150 pounds for a minimum period of five (5) seconds without an increase in length of more than 0.005 inch after release of load.

4.6.12 Static load. The complete oxygen system shall be installed as in a container installation in accordance with Drawing 67E25232 and the following tests performed. After completion of the following tests, the oxygen system shall be subjected to and pass the quality conformance inspection specified in table I specified herein.

4.6.12.1 Oxygen system-to-man disconnect. A direct pull of 150 pounds shall be applied to the personal leads half of the oxygen system-to-man disconnect, perpendicular to the aft compartment lid for a period of ten (10) seconds without damage.

4.6.12.2 Aircraft-to-oxygen system disconnect. A direct pull of 300 pounds shall be applied to the aircraft-to-oxygen system disconnect, perpendicular to the parting line for a minimum period of ten (10) seconds without damage.

4.6.12.3 Disconnect cable. The lanyard assembly which is a component part of the aircraft-to-oxygen system disconnect shall be loaded to 500 pounds for a minimum of two (2) seconds without damage.

4.6.12.4 Hose fittings. The fittings on the 70 psi hoses shall be subjected to a minimum pull of not less than 125 pounds, maintained for not less than thirty (30) seconds without damage.

4.6.13 High temperature operation. The oxygen system shall be subjected to high temperature tests in accordance with procedure II of MIL-STD-810. The high temperature test shall be conducted at 71°C (160°F) for 12 hours. The system shall be subjected to tests specified in 4.6.2, 4.6.3, and 4.6.4 at the end of the high temperature test and while still at the test temperature. The oxygen system shall then be returned to standard atmospheric conditions and retested in accordance with the quality conformance inspection specified in table I.

4.6.14 Low temperature operation. The oxygen system shall be subjected to low temperature test in accordance with procedure I of MIL-STD-810. All of the low temperature tests shall be conducted at -54°C (-65°F) while the system is subjected to the quality conformance inspection specified in table I.

4.6.15 Low temperature exposure. The oxygen system shall be subjected to a temperature of minus 85 degrees, plus or minus 5 degrees, fahrenheit, for not less than 48 hours. After the 48 hour period, the oxygen system shall be allowed to return to room temperature and then shall be subjected to the quality conformance inspection of table I & II.

4.6.16 Humidity. The aircraft-to-oxygen system disconnect and the oxygen system-to-man disconnect shall be subjected to humidity tests in accordance with procedure I of MIL-STD-810. Upon completion of the humidity tests, both disconnects shall be returned to standard conditions and dried by air blast. Both disconnects shall then be subjected to and meet within 1 hour the quality conformance inspection of table I. Humidity tests performed under the sampling plan tests shall be done in accordance with procedure I of MIL-STD-810 except that it shall be for 72 hours.

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4.6.17 Overpressure. All hose fittings and all other parts where gas pressure is contained, shall be subjected to a minimum of 2-1/2 times the normal working pressure for not less than 5 minutes. The hose fittings and all other parts where gas pressure is contained shall then be subjected to and meet the quality conformance inspection of table I.

4.6.18 Vibration. The complete oxygen system shall be installed as in a container installation. The complete assembly shall be mounted on a vibration stand for six hours on each of three axes - two horizontal and one vertical - operating at a frequency ranging from 500 to 2500 cycles per minute, with a double amplitude between 0.034 and 0.036 inch. The oxygen system shall be subjected to and meet the requirements of the quality conformance inspection of table I & II after the vibration test.

4.6.19 Flexure-disconnect cable. The cable with attachment hardware shall be cycled 1000 times simulating seat adjustment. The cable shall not incorporate any flexures which will affect the life of the cable. Also, the attaching hardware shall not create sharp bends, wear points, etc., in the cable when exposed to expected service conditions including maintenance.

4.6.20 Reliability tests. The complete oxygen system shall be installed as in a container installation (see AF drawing 67E25232) with the emergency oxygen system charged to 1800 psi, the aircraft supply at 100 psi, the pressure reducer set and the suit and helmet connectors blocked off. The disconnect cable shall be pulled within 10 degrees of the line of travel of the parting disconnect at a force of 40 pounds, plus or minus 20 pounds and at a speed of 40 feet per second, plus or minus 5 feet per second. Successive tests shall be conducted manually. Failure of any one of the following operations shall be cause for rejection:

a. Separation of aircraft to oxygen system disconnect.

b. Actuation of the emergency oxygen system.

c. Release of the kit-to-man disconnect by a cable pull simulating the kit handle pull of not more than 5 pounds.

No parts may be replaced as preventive maintenance during reliability testing.

4.6.21 Longevity. The complete oxygen system shall be subjected to 600 successive simulated cycles. A cycle shall consist of one operation as specified in paragraph 3.5.1 subparagraph a(2). After 600 cycles are completed, the requirements of 3.5.1.a(2) shall be met.

4.7 Preservation and packaging. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspections required, as listed in Table III, MIL-P-116 for the method of preservation being accomplished.

5. PACKAGING

5.1 Preservation-packaging. Preservation packaging shall be in accordance with MIL-STD-794, Level A or C as specified by the contracting activity.

5.1.1 Level A. Each oxygen container shall be preserved and packaged in accordance with method 1A of MIL-P-116, without the application of a contact

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preservative. Cleaning shall be accomplished with immersion in solvent. The container shall be placed within the water-vaporproof enclosure required to accomplish this method and sealed.

5.1.2 Level C. Each unit shall be packaged in a manner that will prevent physical damage or deterioration during transit from the supply source to the first receiving activity for immediate use. The package shall be capable of precluding contamination that might impair the use or efficiency of the containers. The contractor's commerical packaging may be utilized when it conforms to the criteria stated herein.

5.2 Packing. Unless otherwise specified by the contracting activity, the container shall be provided packing which will affort adequate protection against damage during shipment, handling, intermediate storage and work-wide redistribution.

5.2.1 Level A. Containers packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style RSC, grade V3s or V3c. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistant with the protection required, and contain identical quantities. Closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60. Reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding.

5.2.2 Level B. Containers packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to style RSC, class domestic, minimum grade 200 of PPP-B-636. Containers shall be of uniform shape and size and contain identical quantities. The container closure shall be accomplished in accordance with method II, 30.3.1.2 of the appendix of PPP-B-636.

5.2.3 Level C. The packaged containers shall be packed in shipping containers in a manner that will affort adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping container.

> "Oxygen Components - All oil, grease, shop residue or other contaminants have been removed.
>
> Do not open until ready for use."

6. NOTES

6.1 Intended use. The oxygen system covered by this specification is intended for use in survival container kits by aircrew members under flight conditions in ejection-seat-equipped, high-altitude aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents should specify the following:

a. Applicable drawing number.

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b. Title, number and date of this specification.

c. First article testing requirements (see 4.4).

d. Whether intended for full pressure suit, partial pressure suit, bladder suit or oxygen mask application.

e. Selection of applicable levels of preservation, packaging, and packing (see section 5).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
4.4.1	First article inspection test report	DI-T-2072	--------

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. (ref. 4.4) When first article inspection is required, the item should be a first article sample or may be a standard production item from the contractors inventory. The first article shall consist of two complete assemblies. A first article test report shall be prepared. The contracting officer will include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. The contracting activity reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Waiver of first article inspection does not relieve the contractor of meeting the quality conformance requirements.

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I Group A quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Examination of product	3.2, 3.3, 3.5, 3.9, 3.10	4.6.1
Kit-to-man disconnect	3.4.4	4.6.2
Leakage, kit-to-man disconnect	3.4.4	4.6.3
Operation "green apple"	3.4.9	4.6.4
Aircraft-to-oxygen system disconnect	3.4.6	4.6.5
Leakage-oxygen systems	3.4.2, 3.4.10	4.6.7
Electric continuity	3.4.5, 3.4.11, 3.4.12, 3.9.3	4.6.8
Dielectric strength	3.4.5, 3.4.6, 3.4.11, 3.9.3	4.6.9
Leakage, oxygen manifold	3.4.1, 3.4.3	4.6.10
Proof load, disconnect cable	3.4.9	4.6.11

TABLE II Groupe B quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Static load		4.6.12
Oxygen system-to-man disconnect	3.4.4	4.6.12.1
Aircraft-to-oxygen system disconnect	3.4.6	4.6.12.2
Disconnect cable	3.4.6, 3.9.6	4.6.12.3
Hose fittings	3.4.4, 3.4.6, 3.4.15	4.6.12.4
High temperature operation	3.4.2, 3.4.3, 3.4.4, Table I	4.6.13
Low temperature operation	Table I	4.6.14
Humidity	3.4.4, 3.4.6, Table I	4.6.16
Overpressure	Table I	4.6.17

TABLE III Group C first article insection.

Inspection	Requirement paragraph	Test paragraph
Low temperature exposure	Table I and II	4.6.15
Vibration	Table I and II	4.6.18
Flexure-disconnect cable	3.4.9	4.6.19
Reliability tests		4.6.20
Longevity	3.5.1	4.6.21

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a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

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